

Khava Jabbareli

Assignment 2

- The first display -

My number: 452736

	$X=0$	$X=1$
ABC	$A_N B_N C_N$	$A_N B_N C_N$
0	000	xxx
1	001	xxx
2	010	111
3	011	110
4	100	101
5	101	010
6	110	100
7	111	011

$$A_N = \Phi A_B$$

0	0	0	0
1	1	1	1
1	0	0	1
x	x	1	x

Groupings: a (rows 1, 2), b (columns 3, 4), c (columns 1, 2), x (rows 3, 4)

$$\Phi_A = a x' + x a' + \cancel{x c'}$$

$$\Phi_A = (a \oplus x) + \cancel{x c'}$$

0	0	1	1
0	0	1	1
0	1	1	0
x	x	1	1

Groupings: a (rows 1, 2), b (columns 3, 4), c (columns 3, 4), x (rows 3, 4)

We transfer $B_v \rightarrow T_b$

T_b

0	0	0	0
0	0	0	0
0	1	0	0
x	x	0	0

Dimensions: $a=2, b=4, c=4$

$$T_b = xcb' + abc'x$$

G_v

0	1	1	0
0	1	1	0
1	0	1	0
x	x	0	1

Dimensions: $a=2, b=4, c=4$

$\rightarrow J_c$

J_c

0	x	x	0
0	x	x	0
1	x	x	0
x	x	x	1

Dimensions: $a=2, b=4, c=4$

K_c

x	0	0	x
x	0	0	x
x	1	0	x
x	x	1	x

Dimensions: $a=2, b=4, c=4$

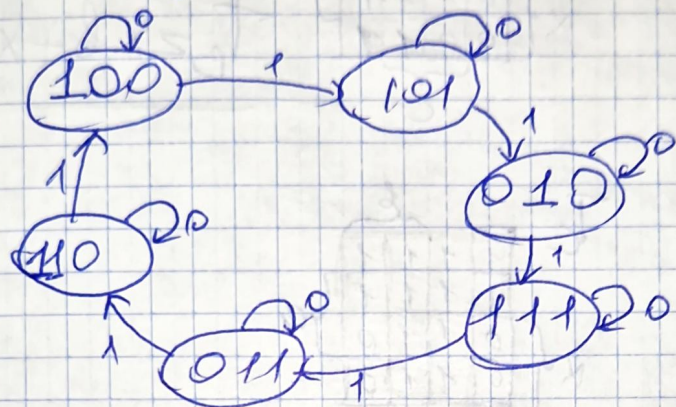
$$J_c = xb' + xa'$$

$$J_c = x(a' + b')$$

$$K_c = xa' + xb'$$

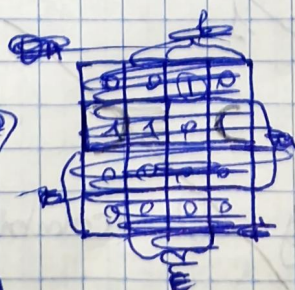
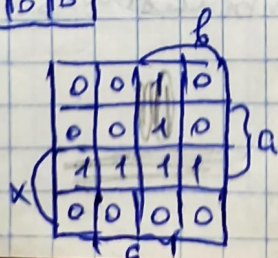
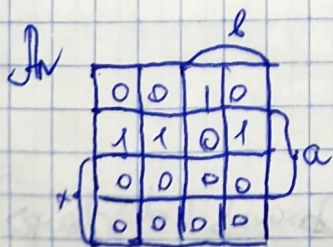
$$K_c = x(a' + b')$$

Working principle, FSM. for 1st display



The second display (count)

	x=0	x=1
ABC	A _x B _x C _x	A _x B _x C _x
000	001	000
001	010	000
010	011	000
011	100	000
100	101	000
101	110	000
110	111	000
111	000	000

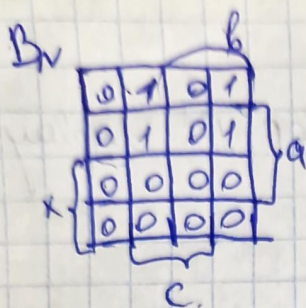


~~$$A = x'ab + x'ac$$~~

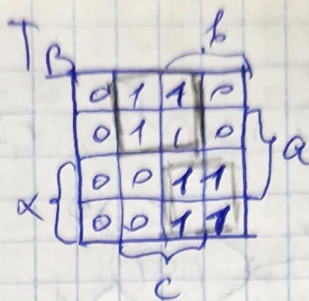
~~stable~~
complex, big up.

Trayer T ff

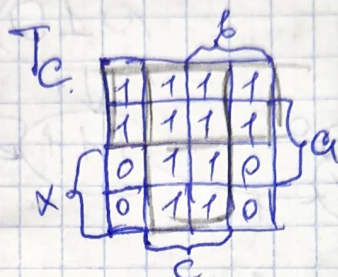
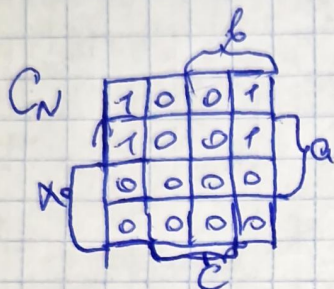
$$TA = ax + bcx'$$



$\Rightarrow$

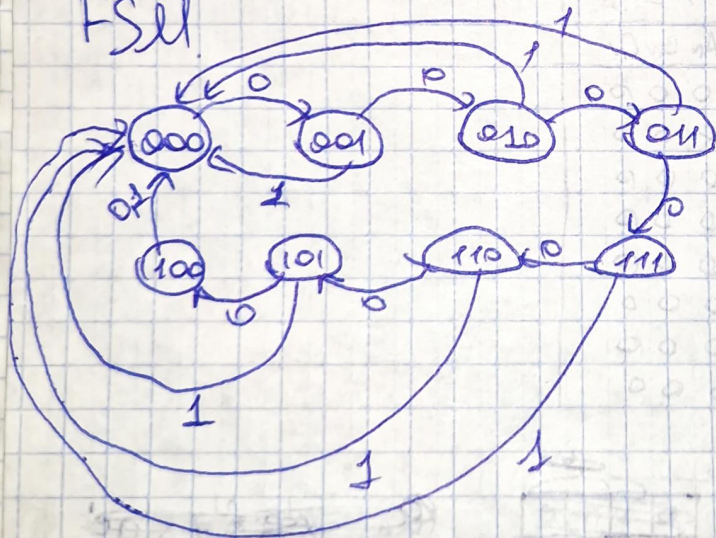


$$T_B = x'b + x'c$$



$$T_C = c + x'$$

FSE



Connecting 2-combinations (displays)

